

Problem 9

Prove the general formulas for powers of sine:

$$\begin{aligned}
 1^\circ \quad \sin^{2n} x &= \frac{(-1)^n}{2^{2n-1}} \left[\cos 2nx - 2n \cos(2n-2)x + \frac{2n(2n-1)}{1 \cdot 2} \cos(2n-4)x + \cdots + \frac{(-1)^n}{2} \cdot \frac{2n(2n-1) \cdots (n+1)}{n!} \right] \\
 2^\circ \quad \sin^{2n+1} x &= \frac{(-1)^n}{2^{2n}} \left[\sin(2n+1)x - (2n+1) \sin(2n-1)x \right. \\
 &\quad + \frac{(2n+1)2n}{1 \cdot 2} \sin(2n-3)x + \cdots \\
 &\quad \left. + (-1)^n \frac{(2n+1)2n \cdots (n+2)}{n!} \sin x \right]
 \end{aligned}$$

Solution

We will use Euler's formula and the binomial theorem to derive these general formulas.

Key Formula

Using Euler's formula:

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

Therefore:

$$\sin^n x = \left(\frac{e^{ix} - e^{-ix}}{2i} \right)^n = \frac{(e^{ix} - e^{-ix})^n}{(2i)^n}$$

Part 1^o: Even Powers $\sin^{2n} x$

Step 1: Setup

$$\sin^{2n} x = \left(\frac{e^{ix} - e^{-ix}}{2i} \right)^{2n} = \frac{(e^{ix} - e^{-ix})^{2n}}{(2i)^{2n}}$$

Note that $(2i)^{2n} = 2^{2n} \cdot i^{2n} = 2^{2n} \cdot (i^2)^n = 2^{2n} \cdot (-1)^n$.

$$\sin^{2n} x = \frac{(e^{ix} - e^{-ix})^{2n}}{2^{2n}(-1)^n}$$

Step 2: Binomial Expansion

$$\begin{aligned}
 (e^{ix} - e^{-ix})^{2n} &= \sum_{k=0}^{2n} \binom{2n}{k} (e^{ix})^{2n-k} (-e^{-ix})^k \\
 &= \sum_{k=0}^{2n} \binom{2n}{k} (-1)^k e^{i(2n-2k)x}
 \end{aligned}$$

Step 3: Pair Symmetric Terms

For even powers, we pair terms with exponents $(2n-2k)$ and $-(2n-2k)$:

$$e^{i(2n-2k)x} + e^{-i(2n-2k)x} = 2 \cos((2n-2k)x)$$

The middle term (when $k = n$) gives a constant:

$$\binom{2n}{n} (-1)^n$$

Step 4: Collect Terms

After pairing and simplifying:

$$(e^{ix} - e^{-ix})^{2n} = 2 \sum_{k=0}^{n-1} \binom{2n}{k} (-1)^k \cos((2n-2k)x) + \binom{2n}{n} (-1)^n$$

Step 5: Final Form

$$\begin{aligned} \sin^{2n} x &= \frac{1}{2^{2n}(-1)^n} \left[2 \sum_{k=0}^{n-1} \binom{2n}{k} (-1)^k \cos((2n-2k)x) + \binom{2n}{n} (-1)^n \right] \\ &= \frac{(-1)^n}{2^{2n-1}} \left[\sum_{k=0}^{n-1} \binom{2n}{k} (-1)^k \cos((2n-2k)x) + \frac{1}{2} \binom{2n}{n} (-1)^n \right] \end{aligned}$$

Writing out the terms explicitly:

$$\boxed{\sin^{2n} x = \frac{(-1)^n}{2^{2n-1}} \left[\cos 2nx - 2n \cos(2n-2)x + \frac{2n(2n-1)}{2} \cos(2n-4)x + \cdots + \frac{(-1)^n}{2} \binom{2n}{n} \right]}$$

Part 2°: Odd Powers $\sin^{2n+1} x$ **Step 1: Setup**

$$\sin^{2n+1} x = \left(\frac{e^{ix} - e^{-ix}}{2i} \right)^{2n+1} = \frac{(e^{ix} - e^{-ix})^{2n+1}}{(2i)^{2n+1}}$$

Note that $(2i)^{2n+1} = 2^{2n+1} \cdot i^{2n+1} = 2^{2n+1} \cdot i^{2n} \cdot i = 2^{2n+1} \cdot (-1)^n \cdot i$.

$$\sin^{2n+1} x = \frac{(e^{ix} - e^{-ix})^{2n+1}}{2^{2n+1}(-1)^n i}$$

Step 2: Binomial Expansion

$$(e^{ix} - e^{-ix})^{2n+1} = \sum_{k=0}^{2n+1} \binom{2n+1}{k} (-1)^k e^{i(2n+1-2k)x}$$

Step 3: Pair Terms

For odd powers, all terms pair except none (since $2n+1$ is odd):

$$e^{i(2n+1-2k)x} - e^{-i(2n+1-2k)x} = 2i \sin((2n+1-2k)x)$$

Step 4: Collect Terms

$$(e^{ix} - e^{-ix})^{2n+1} = 2i \sum_{k=0}^n \binom{2n+1}{k} (-1)^k \sin((2n+1-2k)x)$$

Step 5: Final Form

$$\begin{aligned}\sin^{2n+1} x &= \frac{2i \sum_{k=0}^n \binom{2n+1}{k} (-1)^k \sin((2n+1-2k)x)}{2^{2n+1} (-1)^n i} \\ &= \frac{(-1)^n}{2^{2n}} \sum_{k=0}^n \binom{2n+1}{k} (-1)^k \sin((2n+1-2k)x)\end{aligned}$$

Writing out explicitly:

$$\boxed{\sin^{2n+1} x = \frac{(-1)^n}{2^{2n}} \left[\sin(2n+1)x - (2n+1) \sin(2n-1)x + \frac{(2n+1)2n}{2} \sin(2n-3)x + \dots \right]}$$

Verification: Special Cases

For $n = 1$ in formula 2°

$$\sin^3 x = \frac{(-1)^1}{2^2} [\sin 3x - 3 \sin x] = -\frac{1}{4} [\sin 3x - 3 \sin x] = \frac{1}{4} [3 \sin x - \sin 3x]$$

This matches the known formula.

For $n = 2$ in formula 1°

$$\sin^4 x = \frac{(-1)^2}{2^3} \left[\cos 4x - 4 \cos 2x + \frac{4 \cdot 3}{2} \right] = \frac{1}{8} [\cos 4x - 4 \cos 2x + 6]$$

This can be verified using standard identities.

Final Answer

The general formulas for powers of sine are:

$$\boxed{\begin{aligned}\sin^{2n} x &= \frac{(-1)^n}{2^{2n-1}} \left[\sum_{k=0}^{n-1} \binom{2n}{k} (-1)^k \cos(2n-2k)x + \frac{(-1)^n}{2} \binom{2n}{n} \right] \\ \sin^{2n+1} x &= \frac{(-1)^n}{2^{2n}} \sum_{k=0}^n \binom{2n+1}{k} (-1)^k \sin((2n+1-2k)x)\end{aligned}}$$

These formulas express any integer power of sine as a linear combination of cosines (for even powers) or sines (for odd powers) of multiple angles.